

Mustafa Emir Kavçakar

Energy Systems Engineering Student

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SUMMARY

Energy Systems Engineering student with field experience at a cogeneration plant covering gas turbines, HRSG systems and electrical protection. Hands-on experience in EV charging station commissioning, CFD analysis and 3D-printed prototyping. Founder and operator of a 3D printing business.

EDUCATION

B.Sc. Energy Systems Engineering, Yaşar University, İzmir

2023 – 2026

Relevant courses: Power Conversion Systems, Engineering Project Management, Thermodynamics, Electromechanics, Generative AI for Engineers

EXPERIENCE

Engineering Intern, Desa Elektrik Üretim A.Ş. (Yaşar Holding), İzmir

Summer 2026

- Supported operation of Solar Taurus 60 gas turbines and HRSG/boiler systems at a cogeneration plant.
- Took part in commissioning a 22 kW three-phase EV charging station (Sinexcel) and troubleshot OCPP 1.6 connectivity over LTE.
- Participated in diesel generator testing (Kipor KD6142ZL).
- Worked with electrical protection equipment including MCCB panels, current transformers and distribution blocks.
- Studied field instrumentation: Siemens SITRANS P pressure transmitters, Endress+Hauser iTHERM temperature sensors.
- Reviewed plant operating economics.

Founder, MeKa 3D, İzmir

Present

- Run a 3D printing business: model sculpting, multicolor FDM production, pricing and online sales.
- Built the e-commerce website (3dmeka.com) with Shopier payments, product schema and Google Merchant Center setup.

PROJECTS

Analysis and Design of Cross-Flow Air-to-Air Heat Exchangers (Graduation Project, YU-COOP)

- Industry-partnered team project; CFD analysis in ANSYS Fluent.

Automatic-Switchover UPS for a 3D Printer (Personal Project)

- Designed an offline relay-based UPS using a 12 V 9 Ah battery and inverter, including DC-rail capacitor sizing for transfer time.

TeknoSerious (Tech Blog)

- Publish a Turkish technology news blog; developed custom Blogger themes and a Python/Pillow cover image pipeline.

SKILLS

Technical: Gas turbine and HRSG operation, electrical protection, EV charging systems, OCPP, CFD, 3D printing and prototyping

Software: ANSYS Fluent, Python, HTML/CSS/JavaScript, Bambu Studio / Orca Slicer, Git and GitHub Pages

Languages: Turkish (native), English (B2)